AI Image Generator

12. Operations and Troubleshooting

Version 2.0
22 October 2026

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Platform: NVIDIA GeForce RTX 3090, Windows 11, Python 3.12

Z-Image Turbo, Stable Diffusion XL and its best fine-tunes, FLUX.1, LoRAs and Real-ESRGAN, served locally with Gradio in the Frazile AI studio UI, with per-image GPU telemetry recorded in SQLite and a live Admin panel.

12. Operations and Troubleshooting

12.1 Starting and stopping

Action	How
Start	Double-click run-app.bat, or `.\.venv\Scripts\python.exe app.py` in the project folder; or `docker compose up`
Confirm ready	Console shows "SDXL base 1.0 ready in N s", "Upscaler ready", "Running on local URL: http://127.0.0.1:7860"
Stop	Ctrl+C in the console, or close the window; `docker compose down` for the container
Change port or address	Set GRADIO_SERVER_PORT or GRADIO_SERVER_NAME before starting; FRAZILE_NO_BROWSER=1 suppresses the browser tab
Expose on the LAN	GRADIO_SERVER_NAME=0.0.0.0. There is no authentication, so only do this on a trusted network

12.2 Logs

The app prints one line per loaded model, per generated image and per upscale, for example:

```
Loading Juggernaut XL v9 ...
Juggernaut XL v9 ready in 26.3s  (scheduler: DPM++ 2M Karras)
#12 juggernaut saved 20261022-101040_seed1746058488.png  832x1216  8.0s  peak 13.1 GB  56.0 C
#12 upscaled -> 1463x2560 in 3.56s
```

To keep a log file, redirect: `.\.venv\Scripts\python.exe -u app.py > app.log 2>&1`. Inside Docker use `docker compose logs -f`.

12.3 Known issues encountered during development and their fixes

Symptom	Cause	Fix applied
AttributeError: 'StableDiffusionXLPipeline' object has no attribute 'enable_vae_slicing'	diffusers 0.41 removed the pipeline-level helper	Call pipe.vae.enable_slicing() instead
Parser ignored "seed 42" and misread "21:9 steps 45" as 9 steps	Directive removal used offsets from a stale lowercase copy of the prompt	Rewrote removal to search and cut the live prompt string each time
Regex word boundaries silently vanished	A patch script wrote "\b" through a normal Python string, turning it into a backspace	Patches now use raw strings or the file is rewritten whole
compel: 'EmbeddingsProviderMulti' object has no attribute 'empty_z'	compel's pad_conditioning_tensors_to_same_length does not support SDXL's two encoders, so a long prompt with a short negative failed	models.py pads the shorter embedding with chunks of the empty-prompt embedding
Theme build failed: Base.set() got an unexpected keyword argument 'color_accent_dark'	Not every Gradio theme variable has a dark-mode twin	build_theme() filters keys against the signature of Base.set()
Admin refresh took 3 s every time	Walking the 39 GB Hugging Face cache and every process for thread counts	Folder sizes cached 60 s, thread count cached 30 s; refresh is 0.7 s
Segmentation fault while generating with Z-Image Turbo	enable_model_cpu_offload() with the 23 GB fp32 transformer on a 16 GB RAM machine	Transformer loaded shard by shard onto the GPU in bf16 with device_map; text encoder in 8-bit; bf16 copy cached under models/cache/
Z-Image generation took 78 to 128 s	The web app was still holding 7 GB of VRAM during the benchmark, so Windows paged GPU memory	Stop other GPU processes before benchmarking; with the GPU free the same images take 8.6 to 9.8 s
"strict" reported higher guidance for FLUX	Fidelity words were noted before the model was known	apply_defaults() notes that the model has no guidance scale

Symptom	Cause	Fix applied
Space before a comma after a removed tag ("cat , quick")	Tag removal left the separator behind	Tidy step also collapses whitespace before commas
pip could not write pymupdf.exe in C:\Python312\Scripts	Permission problem in the system Python's Scripts folder	Not a project issue; the venv is unaffected. The documentation tooling used the system Python only for one-off edits
UnicodeEncodeError when printing reports from scripts	Windows console code page cp1252 cannot print the arrow and dot characters	Set PYTHONIOENCODING=utf-8 when printing reports; the browser is unaffected
torchvision installed as the CPU build	pip pulled torchvision from PyPI before the CUDA index was consulted	Reinstalled from the cu126 index; requirements.txt documents the order

12.4 Troubleshooting guide

Problem	Check	Resolution
"CUDA GPU not available" at startup	`python -c "import torch; print(torch.version.cuda)"` prints None	Reinstall torch from the cu126 index inside the virtual environment
Port 7860 already in use	Task Manager shows a python.exe from a previous run	End that process, or set GRADIO_SERVER_PORT
"Unknown model 'x'" in the preview line	Typo, or a checkpoint added without restarting	Use a key from the Models accordion; restart after adding files to models/checkpoints/
Generation waits 30 s before starting	The progress bar says "Loading <model>"	Normal for a model switch. Keep the Model box on one model, or Auto, for consecutive prompts
First use of a model waits minutes	Progress says "downloading ~6.5 GB"	Wait; check the HF cache folder growth
"FLUX.1 [schnell] is gated"	No Hugging Face token stored	Accept the terms on the model page and run huggingface-cli login (section 4.10)
LoRA has no effect and the notes say "not found"	File name does not match the tag	Compare with the LoRA list in the Models accordion; matching ignores case and punctuation
Out of memory during generation	Other GPU programs running; or a card under 12 GB; or Z-Image or FLUX on under 24 GB	Close other GPU apps; use generate.py which offloads to CPU on small cards; Z-Image and FLUX need 24 GB
Z-Image is very slow (over a minute per image)	Another process holds several GB of VRAM; nvidia-smi shows memory near 24 GB	Close it; Z-Image needs about 18 GB free
Telemetry fields show None	nvidia-smi not on PATH	Reinstall the NVIDIA driver, or add C:\Windows\System32 to PATH
Upscaler weights missing	FileNotFoundError naming models/RealESRGAN_x4plus.pth	Download from the Real-ESRGAN v0.1.0 release into models/
Upscale bar does nothing	No generation in this browser session	Generate first; the bar works on the records of the last generation
History tab empty	New database, or generations.db moved	Generate an image; or restore the file next to db.py
Admin panel shows 0 B/s network	First sample	Refresh again; rates need two samples
Browser slow to show an 8K result	30 MB PNG transfer	Expected; open the file from outputs/ directly
docker compose fails with "could not select device driver"	GPU support not enabled in Docker	Install WSL 2 and Docker Desktop (Windows) or nvidia-container-toolkit (Linux)

12.5 Maintenance

- Outputs grow by 1.5 MB per native image and 3 to 31 MB per upscaled image. Prune outputs/ periodically; the History tab will show "not found" for deleted files. The Admin panel shows the folder size.
- Each built-in model is 6.5 GB in the Hugging Face cache. Delete a models--* folder there to reclaim space; it downloads again on next use.
- Back up outputs/, generations.db and models/ together so paths stay valid.
- To reset history, stop the app and delete generations.db; it is recreated empty on next start.
- To update dependencies, upgrade inside the virtual environment and rerun the parser self-test, models.py, sysinfo.py and one generation per family. Pin versions in requirements.txt if a breaking change is found.

12.6 Security notes

- The server listens on localhost only by default and has no authentication. The Docker image binds all interfaces inside the container; the compose file publishes the port on the host, so treat that like a LAN exposure.
- No telemetry or prompts are sent anywhere. Network access is limited to model downloads from Hugging Face and the optional reachability check in the Admin panel, which only opens a TCP connection.
- The Admin panel reveals hostnames, IP addresses, MAC addresses, listening ports and process names. Anyone who can reach the page can read them.
- Prompts are stored verbatim in generations.db. Treat the file with the same care as the images.